

Ideathon Pitch: JanMitra-Kinship Ecosystem

Theme: Progress Without Privilege

One-Sentence Pitch: A solar-powered hardware gateway and peer-to-peer AI mesh that localizes Digital India, protecting rural wealth and accelerating student opportunities without requiring a single byte of cloud data.

1. The Problem: The Connectivity Ceiling

Digital India is built on the cloud, but the "Bottom 50%" live in the shadows.

- **Data Poverty:** Education is now digital, but data is a recurring tax rural students cannot pay.
- **The Scheme Gap:** Over **₹1.2 Lakh Crore** in subsidies goes unclaimed due to "Information Asymmetry."
- **The Trust Deficit:** 1st-generation users are losing life savings to predatory UPI scams—**₹1,700 Crore lost to cybercrime in India in 2024 alone.**

2. The Solution: Hardware-Software Synergy

We move intelligence from the cloud to the **Edge**.

A. The JanMitra Box (ECE Edge)

- **The Hub:** A solar-powered **Raspberry Pi 4** node acting as a "Local Cloud."
- **Storage:** 64GB Industrial SD Card mirroring DigiLocker, NCERT, and Agmarket data.
- **Sync:** When a government official visits town, the box updates via a 5-minute 5G burst, then serves the village offline for weeks.

B. The Kinship Mesh (AIML Brain)

- **Gossip Protocol:** Uses **BLE/WiFi-Direct** (IEEE 802.11s). Phones "handshake" to pass encrypted scholarship and mandi-price packets across the village.
- **Distilled Oracle:** A **4-bit Quantized Llama-3.2 (1B)** model.
- **Constraint-Aware:** Optimized to run on **ARM Cortex-A53** processors (budget ₹8k phones) with < 600MB RAM footprint.

3. Sector-Specific Impact Modules

Agriculture: The Mandi Advantage

- **Yield Simulation:** Voice AI analyzes acreage to cut input waste—**saving ₹5,000/ha.**
- **Market Negotiation:** Offline Mandi trends provide "Sell/Wait" forecasts. AI role-plays bargaining in **local dialects** (Kannada/Hindi) using real-time MSP data.

Education: The Sovereign Student

- **Adaptive Micro-Learning:** AI detects concept-struggle via voice latency and switches to vernacular "Concept Capsules."
- **Scholarship Scout:** Proactive "push" notifications for 200+ offline grants based on the student's encrypted local profile.

Finance: The "Red Alert" Firewall

- **Local NLP:** Scans SMS locally for scam clusters ("KYC Expired "). **Immediate Shield:** Voice warning: "Caution: This message matches fraud patterns. No OTP should be shared."

4. Technical Matrix & Data Safety

Component	Implementation	Sovereign Benefit
Model	Quantized GGUF / INT4	Privacy: Inference is 100% on-device.
Logic	Graph-RAG	Reliability: 0% hallucination on govt rules.
ASR/TTS	Vosk / Whisper-Tiny	Inclusion: Voice-first for low-literacy users.
Security	Asymmetric Encryption	Integrity: All "Public Packets" are signed.

5. Why JanMitra-Kinship Wins...

- **Cost:** Recurring cost is Zero. One JanMitra box serves an entire village.
- **Safety:** Unlike cloud AI, user queries never leave the community mesh, ensuring 100% data sovereignty.
- **Scale:** The network gets stronger as more students/farmers join. If one phone dies, the data is already replicated across the "Kinship" mesh.

Final Elevator Pitch :

"We are not building an app; we are building a **Cognitive Infrastructure**. JanMitra-Kinship ensures that a student's potential or a farmer's wealth isn't dependent on a 4G signal. We bring the 'Harvard Experience' and the 'Mandi Advantage' to the furthest corners of India—all while the signal bars stay at zero."